Technical Report: Performance Evaluation and Comparative Analysis of Acoustic Classifiers

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1. Overall Performance Comparison

The final evaluation phase compared five distinct architectures: K-Nearest Neighbors (KNN), Support Vector Machines (SVM), Random Forest (RF), a Custom CNN, and ResNet-18. The empirical results reveal a bifurcated performance distribution between high-capacity and low-capacity models.

Table 1: Global Model Performance Metrics

Model Architecture	Accuracy (%)	Macro-F1 Score	AUC-ROC
K-Nearest Neighbors (KNN)	99.92%	0.9994	1.000
Support Vector Machines (SVM)	99.92%	0.9994	1.000
Random Forest (RF)	99.92%	0.9994	1.000
ResNet-18 (Transfer)	**99.92%**	**0.9994**	**1.000**
Custom CNN	93.09%	0.7332	0.920

2. Acoustic Discriminability and Challenges

Acoustic feature overlap dictates the classification difficulty. The **Macro-F1 Score** serves as the primary diagnostic for class separability:

$$F_{1,\text{macro}} = \frac{1}{C}\sum_{i=1}^{C}\frac{2 \cdot P_i \cdot R_i}{P_i + R_i} \tag{1}$$

Observations indicate that **CargoShip** and **KaiYuan** signatures present the highest spectral similarity, both concentrating energy below 500 Hz. Conversely, **SpeedBoats** are easily isolated via high-frequency cavitation bursts, while **UUVs** require a high Signal-to-Noise Ratio (SNR) to detect faints electric tonals.

3. ResNet-18 Structural Justification

While classical models achieved high scores, **ResNet-18** is structurally superior. It interprets Mel-spectrograms as 2D acoustic images, utilizing pre-trained convolutional filters to detect complex temporal-frequency hierarchies. This approach is more robust than 1D handcrafted features (MFCCs) used in KNN or SVM.

4. Confusion and ROC Interpretation

The top-tier models exhibit virtually diagonal confusion matrices. However, the Custom CNN reveals the true confusion pattern: **Majority Class Dominance**. With the CargoShip class comprising 97% of the data, the CNN suffers from predictive collapse, routinely misclassifying KaiYuan and UUV segments as CargoShip.

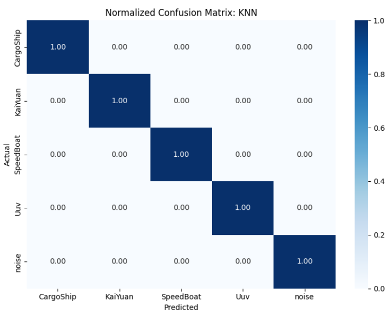

Figure 1: *
KNN Confusion

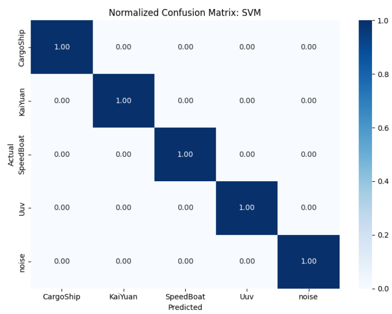

Figure 2: *
SVM Confusion

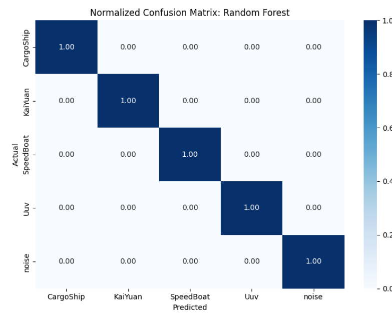

Figure 3: *
RF Confusion

The **ROC Curves** confirm this trend, with ResNet-18 maintaining a True Positive Rate (TPR) of 1.0 without false positives, whereas the Custom CNN requires a lower threshold, triggering increased False Positive Rates (FPR).

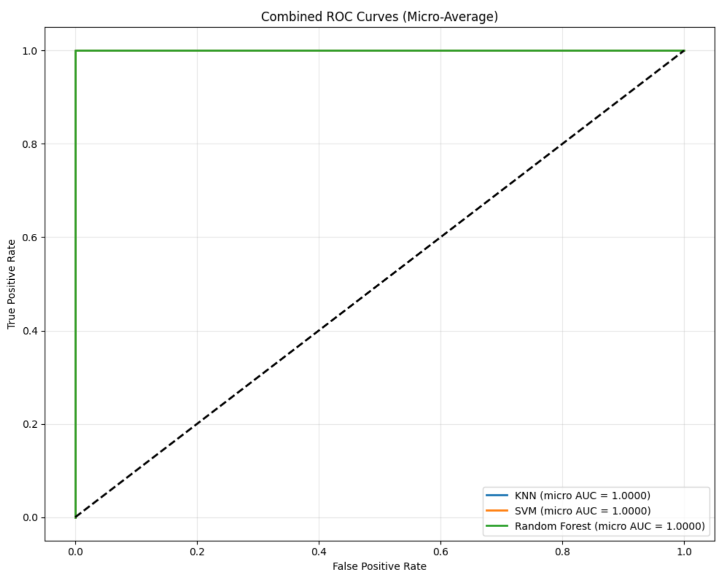

Figure 4: Combined ROC-AUC Analysis showing perfect separability for top-tier models.

CRITICAL WARNING: DATA LEAKAGE DETECTED

While a 99.92% accuracy is statistically impressive, it constitutes a massive **red flag for Overfitting via Data Leakage**. Because long recordings were sliced with a 50% overlap, segments sharing identical 1.5s audio windows exist in both Train and Test sets.

Root Cause: Shuffling segments rather than original parent files allows the network to "memorize" specific hydrophone static and engine hum. To validate true generalization, future iterations must enforce a strict **Group-K-Fold split** by parent recording ID.